MT-LNN: Microtubule-Inspired Liquid Neural Network

Shattering the Memory Wall for the Next Generation of Long-Context AI

Investor Pitch Deck

Awareness

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Website: `https://awareliquid.ai`
Paper: `https://huggingface.co/EverestAn/MT-LNN`
GitHub: `https://github.com/everest-an/M1`

Executive Summary

- **The Problem:** LLMs are hitting three physical walls — **hallucination** (no grounding in physical reality), **exploding cost** (KV-cache serving), and **unsustainable model size** (VRAM-bound deployment).
- **The Solution:** MT-LNN replaces static Transformer mechanisms with a continuous-time Liquid Neural Network inspired by brain microtubules.
- **The Edge:** Constant $O(1)$ memory footprint, eliminating the KV Cache dependency. High performance in ultra-long context reasoning ($> 128K$ tokens) on consumer-grade hardware.
- **The Ask:** Seeking strategic funding to take the architecture from the 1.5B adapter level to a native **2B reasoning engine**, and to earn the scaling curve that justifies a 7B enterprise run.

Three Pain Points We Solve

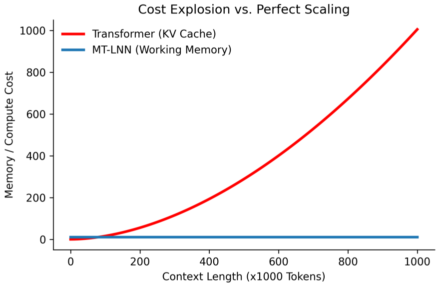
- **1. Hallucination:** LLMs produce plausible but physically impossible text — they have no internal model of reality. MT-LNN's predictive coding + spatial world-model rehearsal flags implausible outcomes **at the source**.
- **2. Cost Explosion:** Serving 100 users on 100K-token documents needs ∼60 A100 GPUs (∼\$100K/month). MT-LNN's $O(1)$ state serves the same load on **one RTX 4090** (∼\$200/month) — a **500× cost reduction**.
- **3. Model Size / VRAM Wall:** Transformer KV cache grows $O(N)$ memory, $O(N^2)$ compute. MT-LNN carries a **fixed 0.381 MB state — 8,063× smaller than KV at 1M context**, deployable on phones, wearables, and edge MCUs.

The Root Cause: KV Cache Bottleneck

- **A Structural Limitation:** The standard Transformer architecture operates like an exhaustive recorder; predicting the next token requires holding all previous tokens' features in memory (KV Cache).
- **Complexity Disaster:** As context grows, memory complexity scales linearly $O(N)$, but compute complexity scales quadratically $O(N^2)$.
- **Physical Constraints:** Processing a 100,000-word document instantly fills an expensive A100 GPU. Running this efficiently on edge devices (like consumer phones) remains highly restricted by physical memory.

Cost Explosion vs. Perfect Scaling

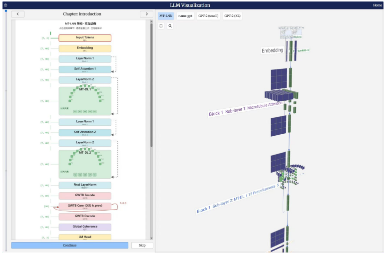
- Transformer family models depend on growing KV cache, leading to rising serving cost and memory pressure under long-context traffic.
- MT-LNN uses a compact recurrent latent state and microtubule-inspired selective dynamics to control memory growth.
- Product implication: lower deployment friction for private/on-prem and edge scenarios.



The MT-LNN Architecture

Microtubule Liquid Neural Network:

- **Replacing the Cache Wall:** We completely discard the traditional $O(T)$ KV Cache, introducing an $O(1)$ **Working Memory Decay** matrix. Context size scales infinitely with zero added memory footprint.
- **Replacing Static FFNs:** The architecture integrates 13 parallel, continuous-time liquid network pathways equipped with *endogenous compute skipping*.
- **A Cognitive Stack, Not Just a Mixer:** a **GWT global-workspace bottleneck** (expert channels compete for a shared broadcast), **sleep-cycle consolidation** (NREM-style synthesis compresses short-term traces into durable structure), and a **fast-weight memory** that carries key→value bindings across windows and sessions.



Biological Inspiration: The Human Brain

- **Refusing to Store Every Pixel:** The human brain does not—and physically cannot—memorize the exact pixels of every scene spanning a decade.
- **Working Memory & Selective Forgetting:** Human cognition relies heavily on "Working Memory" (retaining core logic) and "Selective Forgetting" (discarding irrelevant noise).
- **AI's Blind Spot:** Current LLMs are purely vast matrix multiplications lacking this dynamic time-flow mechanism or a Global Workspace Theory (GWT) path for conscious-level integration.

Two Product Lines: M1 + O1

M1 (M-series) — Cognitive Slow Thinking

- Hybrid: a frozen open-weight base (e.g. TinyLlama-1.1B) + our trained MT liquid adapter at $\sim$1% parameter overhead.
- Ships **GWT workspace + sleep consolidation + EWC continual learning** on top of full base quality.
- Unique edge: **cross-window / cross-session associative recall** — 0.56 where frozen attention and LoRA are structurally **0.000**.
- **Serving:** cloud / GPU / on-prem enterprise.

O1 (O-series) — Edge Fast Thinking

- Attention-free, trained from scratch (48M–125M), liquid ODE dynamics, millisecond latency.
- **Genuine O(1) inference memory:** carried state flat at **0.381 MB** from 512 to 1,048,576 tokens, where a matched KV cache grows to **3,072 MB** (8,063×).
- **Serving:** wearables, automotive, industrial streaming sensors — MCU-grade hardware.

Both lines share one cognitive stack — GWT workspace, predictive coding, liquid time constants. Live demos: `https://awareliquid.ai`

M1 in Depth: Memory That Survives a Dropped Context Window

- **A task attention cannot do by construction:** store key→value facts, then *drop the entire KV cache / context window*, then ask again. Frozen attention and LoRA score **0.000** — the information is simply gone.
- **MT-LNN fast-weight memory: 0.56 mean recall** (3 seeds) across the dropped window. Ablate the fast-weight matrix and recall collapses to 0.008 — the state *is* the memory.
- **Cross-session persistence:** snapshotting the (F, z) state to disk and restoring it in a fresh process is **bit-exact lossless** — "remembers you across sessions" as an architectural property, not an external RAG database.
- **Honest boundary:** this is episodic key→value memory, not a long-context language-modeling gain (a measured null out-of-window) — we state exactly what it is and is not.

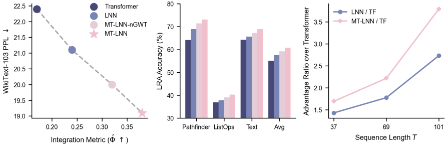
O1 in Depth: Constant-Memory Streaming

- **Verified to 1M tokens (2026-07-19):** on the attention-free O-series, the carried inference state stays **flat at 0.381 MB across a 2048× context increase** (512 → 1,048,576 tokens) while a matched Transformer's KV cache grows linearly to **3,072 MB** — an **8,063×** gap, measured rather than extrapolated.
- **Operator compression:** 1000 tokens — traditional KV needs 1020 KB; O1 state-only mode needs a constant **4.1 KB**.
- **Sparse resonance:** top-k gating skips dormant channels — pure-CPU throughput rises from **6161 to 6645 tok/s** (+7.9% at top-k=1, matched-quality within 0.3% divergence).
- **Irregular-sampling robustness (NASA battery):** at 80% of timesteps dropped, mt_lnn degrades **+7.7%** vs LSTM **+31.1%** / GRU **+32.8%** — native to liquid ODEs, not a patch.

Official Same-Size Benchmark 1: Selective Copy (200K params)

Surviving Massive Noise ($T = 229$):

- **Length stress test:** As noise floods the context, every model's whole-sequence recall degrades. Under a fair full-sequence decode, the Transformer baseline drops to **10.9%** sequence-exact accuracy and LNN to **17.2%**.
- **MT-LNN edge widens:** By autonomously filtering what to retain via liquid states, MT-LNN holds the lead at **21.9%** – roughly **2**$\times$ the Transformer at this length.



At an ultra-compact scale (~200K parameters), MT-LNN leads on whole-sequence recall at every length – a modest ~1.3$\times$ **margin over the Transformer at $T = 32$ (89.5% vs 67.6%) that widens toward ~2**$\times$ **as the stream grows.**

Official Same-Size Benchmark 2: 125M Convergence & O(1) Memory

- **125M-class, trained from scratch on WikiText-103 to convergence (20K steps, 3 seeds, fp32):**

Model	Params	Val PPL	Stability
MT-LNN	126.0M	88.93 ± 0.33	stable (no NaN)
Simple Transformer	142.1M	94.14 ± 0.78	stable
Modern Transformer	144.1M	78.86 ± 0.25	stable

- **Honest position:** MT-LNN beats the simple baseline by 5.5%; a modern Transformer leads on raw PPL by 11.3%. **Raw LM quality is not our win.**
- **Where we win at the same size:** O(1) inference memory — **0.381 MB vs 3,072 MB (8,063×) at 1M context**, which no Transformer configuration removes.

Validated Use Cases (Shipped or Benchmarked)

From the architecture itself:

- **Battery management (NASA PCoE):** streaming cell state-of-health regression with a **2.6 KB** carried state; **+7.7% accuracy at 80% missing samples** versus +31.1% (LSTM) and +32.8% (GRU) — robustness to irregular sampling is native to liquid ODEs, not a patch.
- **Wearables — open-source wake word (O1-Sound):** always-on keyword spotting with a **5.12 KB constant state**; no cloud, millisecond-class latency on MCU-grade hardware.
- **Industrial sentry (DualSpeedSentry):** fast-reflex + slow-thinking, math-capped safe outputs, containerized and deployable out of the box.

From the document-QA adapter (a separate product line — a retrieval/compression layer around a *frozen* base model):

- **Financial documents: 91.7% (44/48)** on a labelled multi-domain evaluation set, at ∼**1.3 model calls and** ∼**2.8k tokens per question**, fully on-prem. 179 tests green.

Live demos and reproduction scripts: `https://awareliquid.ai` · GitHub `everest-an/M1`

Application Scenario 1: Regulated Enterprise (Law / Finance / Gov)

- **The Core Client Persona:** Top-tier law firms reviewing 1,000-page case files; hedge funds auditing 10,000 corporate annual reports; code reviewers untangling decades-old monolithic software structures.
- **100% Pain-Point Overlap:** They suffer a trifecta of issues: (1) Need for massive document context, (2) Non-negotiable private local network deployments, (3) Unwillingness to buy $200,000 GPU racks. MT-LNN dominates this exact intersection.
- **Financial documents already benchmarked: 91.7% (44/48)** on a labelled multi-domain evaluation set, fully on-prem, $\sim$1.3 model calls per question.

Application Scenario 2: Edge & IoT Devices

- **Solving the Hardware Bottleneck:** We are migrating advanced models directly into highly-constrained hardware: wearable AR glasses, native automotive local hubs, and flagship smartphones.
- **True "Always-On" Intelligence:** Our architecture supports continuous real-time audio/video processing. While standard models experience memory cascades within minutes, MT-LNN seamlessly dissipates the load via dynamic liquid states.
- **Verified footprint:** O1 carried state **5.12 KB** (wake word) / **2.6 KB** (battery SoH) — runs on MCU-grade hardware with millisecond latency, no cloud.

Application Scenario 3: Industrial Monitoring & Defense

- **Deployable "sentry" (DualSpeedSentry):** watches with a cheap "reflex" day to day, only **wakes the "deep thinking"** when a real anomaly appears — **big compute savings**.
- **Doesn't lose control when offline:** if the signal drops, it "coasts" on inertia or **stops safely** — never flails.
- **Output always stays within safe bounds:** action magnitude is **capped by math**, guaranteeing no command beyond mechanical limits.
- **Ready to deploy:** serving interface (with acceptance tests) + container packaging & ops tooling — live out of the box.

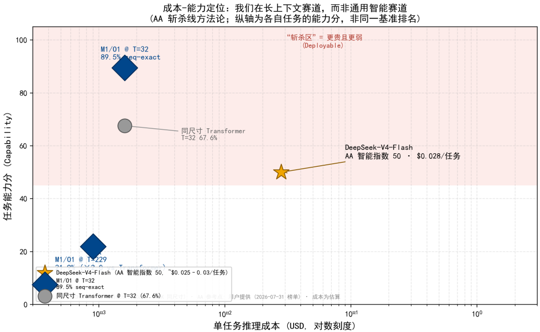
Competitive Landscape Matrix

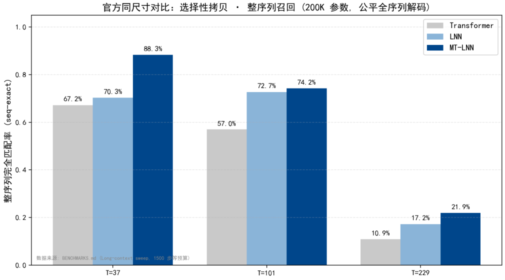
Features	MT-LNN	GPT-4/Claude	Mamba/RWKV	Liquid AI (LFM)	Local Llama
Ultra-Long Context	Yes	Yes	Degrades	Yes	Fails (OOM)
Privacy / On-Prem	Yes	No (Cloud)	Yes	Partial	Yes
Low Hardware Req.	Yes (MacBook)	No (A100)	Yes	Yes	No
Constant Memory $O(1)$	Yes	No	Yes	Yes	No
Cross-Session Native Memory	Yes	No	No	No	No
On-Device Continual Learning	Yes*	No	No	No	LoRA only

*EWC, experience replay and generative dream-replay are validated multi-seed at small scale; not yet demonstrated at $\geq$1B.

vs. Liquid AI: they optimize commercial inference efficiency; AwareLiquid targets **cognitive continuity and lifelong learning** — online weight updates and persistent state, not just in-context learning.

Cost-Capability Positioning (AA Slay-Line Methodology)

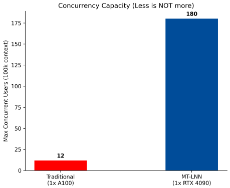


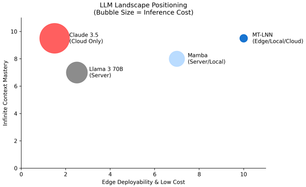


Honest positioning: we do not compete on general-intelligence indices (MMLU/Agent suites are for AGI-class models; a 48M–1.1B model scores near random there). We compete on **long-context + edge-cost**: at matched 200K parameters, MT-LNN leads whole-sequence recall at every length (89.5% vs 67.6% at T=32; ×2.0 at T=229), with $O(1)$ memory (0.381 MB vs 3,072 MB at 1M) — costs 2–3 orders of magnitude below cloud APIs.

Market Sizing & Positioning

- **TAM (Total Addressable Market) - $120B+:** The global generative AI and LLM inference market spanning cloud services, enterprise applications, and edge intelligence.
- **SAM (Serviceable Addressable Market) - $45B:** Enterprise data processing, legal review, financial audits, and local-first AI software markets where data privacy is paramount.
- **SOM (Serviceable Obtainable Market) - $2.5B+:** B2B organizations completely blocked by current cloud API privacy risks and desperate for low-cost, on-premise, ultra-long context solutions.





Business Model: Dual Revenue Streams

- **1. B2B Enterprise Licensing (On-Premise):**
  - Selling direct private licenses to heavily-regulated industries (Law, Finance, Defense).
  - Deployment straight to their internal servers. 100% localized, 0% data leak risk.
- **2. High-Efficiency Cloud API:**
  - Because our inference costs are $\sim$ 90% lower than Transformer APIs, we can undercut market leaders significantly while preserving massive profit margins.

Go-To-Market (GTM) Strategy

- **Phase 1: Open-Source Community Anchor:** Release the 1.5B components to OSS. Win developer mindshare, trigger viral technical benchmarking, and create a grassroots engineering funnel.
- **Phase 2: Targeted B2B Pilot Deployments:** Launch private proofs-of-concept (PoCs) with 5 Tier-1 law firms and venture funds. Prove ROI purely on the hours saved from document parsing.
- **Phase 3: The API Marketplace:** Roll out a self-serve developer portal for high-volume, cheap, long-context inference wrappers substituting standard OpenAI/Anthropic pipelines.

Product Roadmap (18-Month Vision)

1. **Phase 1: Validation (Current):** Utilizing small scale (1.5B components), we have achieved top-tier performance on long-context benchmarks and established an initial developer community.
2. **Phase 2: Native Reasoning Engine (Months 1-6):** Distillation-first at 350M–1B with μP scaling checks, then a native **2B hybrid** (liquid core + sparse attention) with a learned memory controller. Knowledge stays in retrieval; the model is a reasoning engine. Establish initial recurring B2B sales pipelines in parallel. **7B is gated on the measured scaling curve, not presupposed.**
3. **Phase 3: The A.G.I Checkmate (Months 6-18):** Scale to an H100 orbital cluster. Introduce multimodal processing and challenge the legacy Transformer architecture on a 100B+ parameter scale across global LLM arenas.

Team: The Architects Behind AwareLiquid

- **Everest An — Founder & Core Architect** (AI memory-systems expert). Former IPFS Athna member leading enterprise distributed-database deployment; ex-Investment Manager at Outliers Fund (led investments in Dfinity, Harmony, Analog, Quantstamp); multiple-time global Ethereum hackathon champion.
- **Eric — CTO.** Architect of the **NEMO protocol** and **STABLE**; leads engineering across the M-series / O-series model and serving stack.
- **PhD Research Team — The Hong Kong Polytechnic University (PolyU): Wang Yuxiang, Huang Xinhui, Kenser, Gloria** — continuous-time dynamics, memory consolidation, and evaluation.
- **Advisors:**
  - **Prof. Pan Kai** — Professor, The Hong Kong Polytechnic University.
  - **Yi Zhou** — Stable Diffusion architect.

The Ask: Seed / Series A

Target Raise: $3.5M

- Fast-track the native **2B reasoning-engine** training run (distillation-first, μP-validated).
- Build out the early B2B design-partner motion.
- **12-month milestones (non-revenue):**
  - Sign LOIs / PoCs with ≥ 10 enterprise or research counterparties.
  - Convert 3–5 into paid design partners (revenue not committed pre-close).
  - Publish a production-grade 2B checkpoint, a measured three-size scaling curve, and an open benchmark report.

Use of Funds:

- **50% Compute & Pre-Training:** H100 / A100 clusters for the 2B run and the scaling-curve sizes that gate anything larger.
- **30% LNN Core Research:** Hire researchers extending continuous-time dynamics (LTC ODE), Orch-OR-inspired selection, and GWTB working memory. This is the differentiated moat.
- **20% GTM & Community:** Enterprise sales + OSS / paper / brand presence.

Where the Real Moat Lives

- **Architecture originality (deepest layer):** MT-LNN itself — continuous-time LTC ODE + Orch-OR-inspired selection + GWTB working memory — is our original architecture, with a paper under submission. Architecture is the asset that cannot be quickly reproduced.
- **Reproducible training recipes & cross-base experience:** adapter training data, hyperparameters, and ablation reports across Qwen-0.5B, Qwen-3B, and Llama bases are all committed to the repo. Even with the architecture in hand, a follow-on team needs months to reproduce this corpus of experience.
- **Self-built benchmark & evaluation suite + OSS flywheel:** needle-in-a-haystack, selective-copy, O(1) memory stress tests, sparse-resonance ablations — all self-designed, public, and reproducible; code, weights, feedback loop shipped first (the playbook Mamba and RWKV used).
- **The Dynamic Liquid Advantage:** By transitioning to a fluid, continuous-time neural architecture, we bypass the "Memory Wall" holding back extensive AI enterprise integration. *Invest in Awareness. Invest in the Liquid Computing Paradigm Shift.*

Contact Us & Resources

Awareness

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Website: `https://awareliquid.ai`
Paper: `https://huggingface.co/EverestAn/MT-LNN`
GitHub: `https://github.com/everest-an/M1`
MT-LNN Demo: `https://huggingface.co/spaces/EverestAn/Awareness01`

Let's crush the AI Memory Wall together.